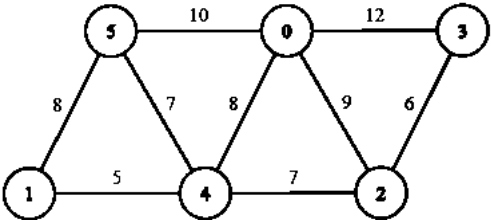
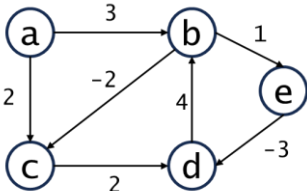
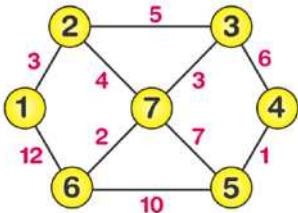




United International University (UIU)
 Dept. of Computer Science and Engineering (CSE)
 Final Exam Year: 2025 Semester: Spring
 Course Code: CSE 2217 Title: Data Structure and Algorithms II
 Marks: 40 Time: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Answer all the questions. All questions are of values indicated on the right-hand margin.

1.	a. Apply Kruskal's Algorithm to find the Maximum Spanning Tree (MST) of the following graph and calculate its total weight. [4]  b. If an undirected graph contains 50 nodes and 1000 edges, which algorithm would you choose between Prim's and Kruskal's to find the minimum spanning tree (MST) of that graph? Justify your answer by mentioning the time complexities of the algorithms. [2] c. “Minimum spanning tree of an undirected graph is unique” - Is this statement true or false? Justify your answer by constructing a graph with 4 nodes. [1]
2.	a. Find the shortest path from ‘a’ to all other vertices in the following directed graph using the appropriate algorithm. [5]  b. If a graph contains a negative weight cycle, why can't we find a valid shortest path from the source to some vertices? Explain with an example. [3] c. Which algorithm you would prefer among BFS, Dijkstra and Bellman-Ford to find the shortest path in the following graph. Justify your answer with mathematical reasoning. [2] 
3.	a. How does path compression heuristic improve the time complexity of find operation in disjoint set data structure? Explain with a proper example. [2]

	b. Table 1 shows the parent array of a Disjoint set (Rooted tree implementation). Perform the following operations sequentially using path compression and union-by-rank heuristic: i. Draw the disjoint set forest. ii. What will be returned by Find-Set(4), and Find-Set(6)? Show details. iii. Redraw the forest after Union(6, 9). iv. Redraw the forest after Union(9, 5). <table border="1"><tr><td>Index</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td></tr><tr><td>Parent</td><td>0</td><td>0</td><td>1</td><td>0</td><td>3</td><td>4</td><td>7</td><td>8</td><td>8</td><td>9</td><td>4</td><td>9</td></tr></table> Table 1: Disjoint set (Rooted tree implementation)	Index	0	1	2	3	4	5	6	7	8	9	10	11	Parent	0	0	1	0	3	4	7	8	8	9	4	9	[2] [2] [1] [1]
Index	0	1	2	3	4	5	6	7	8	9	10	11																
Parent	0	0	1	0	3	4	7	8	8	9	4	9																
4.	a. When would you choose a Direct Address Table over a Hash Table using Chaining? Explain with an example. b. Consider the hash function: $h(k, i) = (h'(k) + i^2) \bmod 13$ where, $h'(k) = (2k + 1) \bmod 13$ By proper calculation, redraw Table 2 and illustrate the following sequence of operations, including accurate hash value calculations. i. Insert 38 ii. Insert 51 iii. Delete 19 and replace with NIL iv. Search 51 v. If your search of 51 fails despite being present at the table, what might be the possible reason? Explain how you can modify the delete operation to prevent this from happening. Currently, the table contains the following: <table border="1"><tr><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr><tr><td>19</td><td>32</td><td>NIL</td><td>NIL</td><td>21</td><td>NIL</td><td>NIL</td><td>42</td><td>NIL</td><td>NIL</td><td>NIL</td><td>70</td><td>NIL</td></tr></table> Table 2: A hash table	0	1	2	3	4	5	6	7	8	9	10	11	12	19	32	NIL	NIL	21	NIL	NIL	42	NIL	NIL	NIL	70	NIL	[2] [1] [1] [1] [1] [2]
0	1	2	3	4	5	6	7	8	9	10	11	12																
19	32	NIL	NIL	21	NIL	NIL	42	NIL	NIL	NIL	70	NIL																
5.	a. What is a spurious hit in the Rabin-Karp algorithm? Explain with an example. b. You are working on a search feature for an e-book reader application. One of the users wants to find whether a specific word segment exists in a book's paragraph. The paragraph contains the text: "sunbeam". The user wants to search for the pattern: "eam". You need to efficiently identify all the occurrences of the pattern within the text. The hash code for each character is given as follows: <table border="1"><tr><td>character</td><td>S</td><td>U</td><td>N</td><td>B</td><td>E</td><td>A</td><td>M</td></tr><tr><td>code</td><td>9</td><td>5</td><td>3</td><td>2</td><td>7</td><td>1</td><td>4</td></tr></table> The hash function is defined as: $hash(s) = (s[m - 1] * d^{m-1} + s[m - 2] * d^{m-2} + \dots + s[1] * d^1 + s[0] * d^0) \bmod q$ where, m = length of the string, d = 7, q = 29 Show the indices of both valid hits and spurious hits (if any) using the aforementioned hash function with detailed simulation.	character	S	U	N	B	E	A	M	code	9	5	3	2	7	1	4	[2] [5]										
character	S	U	N	B	E	A	M																					
code	9	5	3	2	7	1	4																					